

**PROJECT REPORT
ON
COLLEGE PLACEMENT MANAGEMENT SYSTEM**

Carried Out at



**CENTRE FOR DEVELOPMENT OF ADVANCED
COMPUTING
ELECTRONIC CITY, BANGALORE**

UNDER THE SUPERVISION OF

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**PG CERTIFICATE PROGRAMME IN
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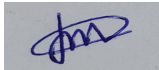
Candidate's Declaration

We hereby certify that the work being presented in the report entitled **College Placement Management System**, in the partial fulfillment of the requirements for the award of **PG Certificate Programme** and submitted in the department of **Advance Computing** of the C-DAC Bangalore, is an authentic record of our work carried out during the **21st April 2026 – 30th July 2026** under the supervision **Mr. Divyansh Verma (Project Engineer)**, C-DAC Bangalore.

The matter presented in the report has not been submitted by me for the award of any degree of this or any other Institute/University.



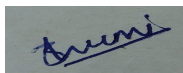
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CERTIFICATE

This is to certify that the project report entitled “**College Placement Manage System**” is a Bonafide record of work carried out by Nakul Mishra, Megha Verma, Saransh Pathak and Bhumika Chandravanshi under my supervision and guidance, in partial fulfilment of the requirements for the award of **PGCP-AC, C-DAC Bangalore**.

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ACKNOWLEDGMENT

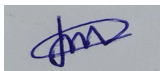
We would like to express our sincere appreciation to everyone who contributed, directly or indirectly, to the successful completion of this project.

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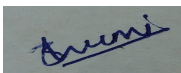
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ABSTRACT

The **College Placement Management System** is a web-based application developed to streamline and automate the campus placement process for educational institutions. Traditional placement management often involves manual handling of student records, company registrations, eligibility verification, job applications, interview scheduling, and placement reporting. These manual processes are time-consuming, prone to errors, and difficult to manage as the number of students and recruiters increases. The proposed system addresses these challenges by providing a centralized, secure, and role-based platform that connects students, recruiters, placement administrators, and super administrators throughout the recruitment lifecycle.

The application is developed using React for the frontend, Spring Boot for the backend, MySQL as the relational database management system, and JWT-based authentication for secure access control. The system follows a modular three-tier architecture consisting of presentation, application, and data layers, ensuring scalability, maintainability, and ease of future enhancements. Major functionalities include secure authentication, student profile management, recruiter management, placement drive management, automated eligibility verification, online job applications, candidate shortlisting, notification management, and placement analytics, as defined in the project's software requirements.

By automating repetitive administrative tasks and providing real-time access to placement information, the system improves communication among all stakeholders while reducing operational overhead. The system enhances transparency in the recruitment process, minimizes manual errors, and provides comprehensive reports that support data-driven decision-making. Overall, the College Placement Management System serves as an efficient, reliable, and scalable solution for managing modern campus placement activities.

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ABBREVIATIONS & ACRONYMS

Abbreviation	Full Form
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DTO	Data Transfer Object
ER	Entity Relationship
ERD	Entity Relationship Diagram
HR	Human Resources
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
JWT	JSON Web Token
ORM	Object Relational Mapping
PMS	Placement Management System
PRN	Permanent Registration Number
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDD	Software Design Document
SRS	Software Requirements Specification
SQL	Structured Query Language
UI	User Interface
UML	Unified Modelling Language
URL	Uniform Resource Locator

INTRODUCTION

1.1 Background

The placement process plays a vital role in shaping the careers of students and enhancing the reputation of educational institutions. Every year, colleges organize campus recruitment drives where companies visit the institution to recruit eligible students for internships and full-time employment opportunities. Traditionally, these placement activities are managed using manual methods such as spreadsheets, emails, printed notices, and physical documentation. As the number of students, recruiters, and placement drives increases, these manual processes become increasingly difficult to manage, leading to inefficiencies, delays, and a higher possibility of errors.

With the rapid advancement of information technology and the widespread adoption of web-based applications, there is a growing need for an automated system that can efficiently manage the complete placement lifecycle. A centralized placement management platform enables students, placement administrators, recruiters, and system administrators to interact through a single secure system. Such a system not only simplifies administrative tasks but also improves communication, transparency, and accessibility for all stakeholders.

The **College Placement Management System** has been developed as a web-based solution to automate placement-related activities within educational institutions. The system integrates multiple modules including student management, recruiter management, placement drive management, eligibility verification, application tracking, notifications, and reporting into a unified platform. By replacing manual procedures with digital workflows, the system minimizes paperwork, reduces processing time, and improves the overall efficiency of campus recruitment activities. These objectives align with the purpose and scope defined in the Software Requirements Specification (SRS).

1.2 Existing System

In many educational institutions, placement activities are still managed through conventional methods involving spreadsheets, emails, paper records, and multiple standalone applications. Student information is maintained separately, recruiters communicate placement opportunities through emails, and placement officers manually verify student eligibility before allowing them to participate in recruitment drives. Application tracking and placement statistics are often maintained using spreadsheets, making the entire process time-consuming and difficult to manage.

These manual practices introduce several operational challenges. Since information is distributed across different platforms, maintaining consistency becomes difficult. Students may miss important placement announcements due to delayed communication, while recruiters may experience delays in receiving updated applicant information. Manual eligibility verification also increases the possibility of human error, leading to incorrect application approvals or rejections.

Furthermore, generating placement reports and analysing recruitment statistics require considerable manual effort. As the volume of placement activities increases, maintaining accurate records becomes increasingly challenging. These limitations highlight the need for an integrated digital solution

capable of automating placement workflows and providing real-time access to information for all users.

1.3 Problem Statement

Managing campus placement activities manually presents several operational and administrative challenges for educational institutions. The absence of a centralized platform results in fragmented data management, delayed communication, repetitive manual work, and increased chances of human error. Placement administrators are required to manage student records, recruiter information, placement drives, eligibility verification, applications, and placement reports separately, making the recruitment process less efficient.

Students often face difficulties in tracking eligible placement opportunities, monitoring application status, and receiving timely notifications regarding interviews or recruitment results. Recruiters also require an efficient mechanism to publish job opportunities, review student profiles, shortlist candidates, and update recruitment outcomes. Without an integrated system, maintaining transparency throughout the recruitment lifecycle becomes difficult.

Therefore, there is a need for a comprehensive web-based application that automates placement management, simplifies administrative operations, improves communication among stakeholders, and provides secure, role-based access to placement-related information.

1.4 Objectives

The primary objective of the College Placement Management System is to develop a secure, centralized, and user-friendly web application that automates the complete campus placement process. The system aims to reduce manual effort, improve operational efficiency, and provide a seamless experience for students, placement administrators, recruiters, and super administrators.

The specific objectives of the project are:

- To provide secure authentication and role-based access for all users.
- To maintain comprehensive student profiles, including academic details and resumes.
- To enable Placement Administrators to create and manage recruiter accounts and associate recruiters with registered companies..
- To automate the creation and management of placement drives.
- To perform automatic eligibility verification based on predefined criteria such as CGPA, department, graduation year, and company-specific requirements.
- To enable eligible students to apply for placement opportunities online.
- To allow recruiters to review applications, shortlist candidates, schedule interviews, and publish recruitment results.
- To provide timely notifications regarding placement activities and recruitment updates.
- To generate placement reports and analytical insights for administrative decision-making.

These objectives are consistent with the project scope and functional requirements defined in the SRS.

1.5 Scope of the Project

The College Placement Management System is designed to serve educational institutions by providing a centralized platform for managing all major placement activities. The application supports four primary user roles: Super Administrator, Placement Administrator, Recruiter, and Student, each with role-specific functionalities and access privileges.

The system covers the complete placement lifecycle, beginning with student account creation and profile management, followed by recruiter onboarding, placement drive creation, eligibility verification, job application management, candidate shortlisting, interview scheduling, recruitment result publication, and placement reporting. The system also provides notification services to keep users informed about important placement events and application updates.

The application is developed as a web-based solution that can be deployed within the institution's infrastructure and accessed through standard web browsers across multiple operating systems. The modular architecture of the system allows future enhancements and integration with additional services without major modifications. This scope reflects the intended functionality described in the SRS.

1.6 Advantages of the Proposed System

The proposed system offers several advantages over traditional manual placement management methods.

- It centralizes all placement-related information within a single web-based platform.
- It reduces manual paperwork and repetitive administrative tasks.
- It automates eligibility verification based on predefined recruitment criteria.
- It improves communication among students, recruiters, and placement administrators through centralized notifications.
- It provides secure authentication using role-based access control and JWT-based authorization.
- It enables students to maintain updated profiles and apply for eligible placement drives online.
- It allows recruiters to efficiently manage job postings, applicant screening, interview scheduling, and recruitment results.
- It generates reports and placement analytics that support institutional planning and decision-making.
- It improves transparency by allowing students to track their application status throughout the recruitment process.
- It provides a scalable architecture that can accommodate increasing numbers of users and placement activities.

1.7 Project Organization

This report is organized into seven chapters, each covering a specific aspect of the project.

Chapter 1 introduces the project by discussing its background, problem statement, objectives, scope, advantages, and overall organization.

Chapter 2 presents a literature survey of existing placement management systems and related technologies.

Chapter 3 describes the Software Requirement Specification, including functional and non-functional requirements, user roles, and system constraints.

Chapter 4 explains the overall system architecture, architectural components, and technology stack.

Chapter 5 focuses on the system design, including database design, module design, and various system diagrams.

Chapter 6 discusses the implementation of the application, highlighting the technologies used and the functionalities developed.

Finally, **Chapter 7** presents the conclusion of the project along with its future scope and possible enhancements.

LITERATURE SURVEY

2.1 Introduction

The rapid advancement of information technology has significantly transformed the way educational institutions manage their academic and administrative activities. Campus placement is one of the most important functions performed by colleges and universities, as it serves as a bridge between students and recruiting organizations. Traditionally, placement activities were conducted manually using paper records, spreadsheets, email communication, and physical notices. While these methods were adequate for small institutions, they became increasingly inefficient as the number of students, recruiters, and placement drives grew.

To address these challenges, many organizations have introduced digital placement management systems that automate recruitment-related processes. Modern placement systems provide centralized platforms where students can maintain their profiles, recruiters can publish job opportunities, placement administrators can manage recruitment drives, and institutions can generate reports for analysis. These systems improve communication, reduce manual effort, enhance transparency, and ensure better coordination among all stakeholders.

Before designing the proposed **College Placement Management System**, a study of existing placement systems and recruitment platforms was conducted to understand their features, advantages, and limitations. This literature survey helped identify the essential functionalities required in a modern placement management system and provided valuable insights for the development of the proposed solution.

2.2 Traditional Placement Management System

Many educational institutions still rely on traditional methods for managing campus placements. Student records are maintained using spreadsheets or paper documents, while placement announcements are communicated through notice boards, emails, or messaging platforms. Eligibility verification is performed manually by placement administrators based on academic records, and recruiters often receive applicant details through separate files or email attachments.

Although this approach is simple to implement, it presents several operational challenges. Managing large volumes of student information manually increases the possibility of data inconsistency and human error. Students may miss important placement updates due to delayed communication, and administrators spend considerable time verifying eligibility, maintaining records, and generating placement reports. The absence of centralized data also makes it difficult to monitor placement activities effectively and analyse institutional placement performance.

These limitations demonstrate the need for an integrated web-based placement management solution capable of automating the complete recruitment process.

2.3 Existing Online Placement Systems

Several online platforms are widely used by educational institutions and recruiters to facilitate campus recruitment. These systems provide centralized services that simplify placement activities and improve collaboration among students, recruiters, and placement administrators.

Common features available in existing placement systems include:

- Student profile management
- Resume uploading
- Company registration
- Job posting and placement drive creation
- Eligibility verification
- Online application submission
- Candidate shortlisting
- Interview scheduling
- Notification management
- Placement reporting and analytics

These platforms have significantly improved the efficiency of placement management by reducing paperwork and enabling digital communication. However, many commercial systems are designed for large organizations and may include features that are unnecessary for individual educational institutions. Additionally, licensing costs, customization limitations, and dependence on third-party platforms may restrict their adoption by smaller colleges.

2.4 Comparative Study of Existing Systems

A comparative analysis of different placement management approaches was carried out to identify the advantages and limitations of each system.

Feature	Traditional Manual System	Existing Online Placement Systems	Proposed College Placement Management System
Student Records	Manual	Digital	Digital
Resume Management	Manual	Available	Available
Recruiter Management	Limited	Available	Available
Placement Drive Management	Manual	Available	Available
Eligibility Verification	Manual	Automated	Automated
Online Applications	Not Available	Available	Available
Role-Based Access	Not Available	Available	Available
Notifications	Manual	Automated	Automated
Report Generation	Manual	Available	Available
Security	Low	Moderate to High	JWT-Based Authentication
Centralized Database	No	Yes	Yes
Scalability	Limited	High	High

The comparison indicates that the proposed system incorporates the essential capabilities of modern placement systems while providing a centralized, secure, and role-based solution suitable for educational institutions.

2.5 Technologies Studied

The development of the proposed system involved studying several modern technologies commonly used in web application development.

React

React is a JavaScript library used for building responsive and interactive user interfaces. It follows a component-based architecture that enables reusable UI components, efficient rendering, and simplified application development. React provides an improved user experience through dynamic page updates without requiring full page reloads.

Spring Boot

Spring Boot is a Java-based backend framework that simplifies the development of enterprise applications by providing built-in configuration, dependency management, and REST API support. It enables rapid development of scalable and maintainable backend services while integrating seamlessly with database technologies.

MySQL

MySQL is an open-source relational database management system used for storing and managing structured application data. It supports efficient data retrieval, indexing, transactions, and relational integrity, making it suitable for placement management applications involving multiple interconnected entities.

JSON Web Token (JWT)

JWT is a secure authentication mechanism used for implementing stateless authentication in web applications. It enables secure user login, role-based authorization, and protected API access by generating digitally signed authentication tokens.

RESTful APIs

RESTful APIs provide standardized communication between frontend and backend applications using HTTP methods such as GET, POST, PUT, and DELETE. They simplify data exchange, improve scalability, and support modular application development.

2.6 Research Gap

The literature survey revealed that although several placement management solutions exist, many institutions continue to depend on manual processes or partially automated systems. Existing commercial platforms often provide generic solutions that may not fully align with the specific operational requirements of educational institutions. Limited customization, dependency on third-party services, and higher deployment costs can reduce their suitability for smaller organizations.

The proposed **College Placement Management System** addresses these challenges by providing a centralized web-based application specifically designed for institutional placement management. The system integrates secure authentication, student profile management, recruiter management, automated eligibility verification, placement drive management, application tracking, notification services, and reporting functionalities into a single platform. This integrated approach improves efficiency, transparency, and user experience while supporting future scalability and maintainability.

2.7 Chapter Summary

This chapter presented a study of traditional placement management practices, existing online placement systems, and the technologies used for developing the proposed system. A comparative analysis highlighted the limitations of manual systems and the benefits of centralized digital platforms. The findings from this literature survey helped establish the functional requirements and architectural decisions adopted for the **College Placement Management System**, which are discussed in the subsequent chapters.

SOFTWARE REQUIREMENT SPECIFICATION

3.1 Introduction

A Software Requirement Specification (SRS) defines the functional and non-functional requirements of a software system. It acts as a blueprint for the design, development, testing, and maintenance of the application. The SRS ensures that all stakeholders, including developers, project guides, testers, and users, have a common understanding of the system's objectives and expected behaviour.

For the **College Placement Management System**, the SRS specifies the system functionalities, user roles, operational environment, external interfaces, performance expectations, security requirements, and quality standards. The document provides a structured description of how the proposed system should operate and establishes the foundation for the subsequent design and implementation phases.

3.2 Purpose

The primary purpose of the College Placement Management System is to provide a centralized web-based platform that automates campus placement activities within educational institutions. The system eliminates manual handling of placement records and streamlines interactions among students, placement administrators, recruiters, and super administrators.

The application enables secure user authentication, student profile management, recruiter management, placement drive management, eligibility verification, online job applications, notification services, and report generation. Through automation, the system improves operational efficiency, reduces administrative workload, enhances transparency, and supports informed decision-making.

3.3 Intended Audience

The proposed system is designed for multiple stakeholders involved in the placement process.

Super Administrator

The Super Administrator manages the overall system configuration, creates Placement Administrator accounts, assigns permissions, monitors system activities, and oversees administrative reports.

Placement Administrator

The Placement Administrator is responsible for managing students, recruiters, placement drives, eligibility criteria, announcements, applications, and placement reports.

Recruiter

Recruiters or company representatives use the system to manage recruitment activities for the associated company, publish job opportunities, define eligibility requirements, review applicants, shortlist candidates, and update recruitment results.

Student

Students access the system to maintain their profiles, upload resumes, view eligible placement drives, submit applications, track application status, and receive notifications regarding recruitment activities.

Project Guide and Evaluators

The SRS also serves as a reference document for project guides, testers, maintenance teams, and evaluators during project review and validation.

3.4 Scope of the System

The College Placement Management System provides an integrated solution for managing the complete placement lifecycle within an educational institution. The system centralizes placement-related information and enables efficient communication among all stakeholders.

The scope of the application includes:

- Student account management
- Student profile management
- Resume management
- Recruiter and company management
- Placement drive management
- Eligibility verification
- Job application management
- Candidate shortlisting
- Notification management
- Placement reporting and analytics

The application is intended to be deployed within institutional infrastructure and accessed through modern web browsers using authenticated user accounts.

3.5 Overall Description

The proposed system follows a centralized architecture in which all users interact with a single web-based application. Depending on their assigned role, users are provided access to specific modules and functionalities.

The system automates the complete recruitment workflow beginning with student registration by the Placement Administrator, followed by recruiter onboarding, placement drive creation, eligibility verification, job application submission, candidate shortlisting, interview scheduling, result publication, and placement reporting.

Role-based access control ensures that each user can only perform actions relevant to their responsibilities, improving both security and operational efficiency.

3.6 User Roles and Responsibilities

User	Major Responsibilities
Super Administrator	Manage Placement Administrators, configure system settings, monitor activities, generate administrative reports
Placement Administrator	Manage students, recruiters, placement drives, eligibility criteria, reports, and announcements
Recruiter	Maintain company profile, publish jobs, shortlist candidates, update recruitment status
Student	Maintain profile, upload resume, apply for placement drives, track applications, view notifications

These responsibilities define the operational boundaries of each user category and ensure secure role-based access throughout the application.

3.7 Functional Requirements

The functional requirements describe the services provided by the application.

The major functional modules of the proposed system are:

Authentication and Authorization

The system provides secure login, logout, password reset functionality, role-based access control, and JWT-based authentication to ensure that only authorized users can access protected resources.

Student Account Management

Placement Administrators can create and manage student accounts. Students cannot self-register and receive credentials from authorized administrators.

Student Profile Management

Students can update personal information, academic details, skills, certifications, and upload resumes required during recruitment.

Recruiter Management

Placement Administrators can create recruiter accounts, update recruiter information, activate or deactivate accounts, and manage recruiter access.

Placement Drive Management

Placement Administrators can create placement drives, define eligibility criteria, schedule recruitment activities, and publish placement opportunities.

Eligibility Verification

The system automatically verifies eligibility based on CGPA, department, graduation year, and company-specific criteria before allowing students to apply.

Job Application Management

Eligible students can browse available placement drives, submit applications, and monitor application status throughout the recruitment process.

Candidate Shortlisting

Recruiters can review applications, shortlist candidates, update recruitment stages, and publish final recruitment results.

Notification Management

The system generates notifications for placement announcements, interview schedules, application status updates, and recruitment results through both the application and email services.

Reporting and Analytics

Authorized users can generate reports related to students, recruiters, placement statistics, departmental performance, and placement outcomes.

3.8 Non-Functional Requirements

In addition to functional capabilities, the system satisfies several quality attributes.

Performance

- Supports multiple concurrent users.
- Maintains fast response time for standard operations.
- Optimizes database performance through efficient queries.

Security

- JWT-based authentication.
- Role-Based Access Control (RBAC).
- Password encryption.
- Protection against unauthorized access.

Reliability

- Database backup mechanisms.
- Error logging.
- Consistent data integrity.

Usability

- Responsive user interface.

- Multi-browser compatibility.
- Simple navigation and intuitive workflows.

Maintainability

- Modular architecture.
- Standardized REST APIs.
- Well-structured database design.

Scalability

The application architecture allows future expansion with minimal changes to existing modules.

3.9 System Requirements

Server Environment

Component	Technology
Operating System	Windows 11, MacOS 26.5, Ubuntu 26.04
Programming Language	Java 21
Backend Framework	Spring Boot 3
Database	MySQL
Authentication	JWT
Build Tool	Maven
Version Control	Git

Client Environment

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Windows
- Linux
- macOS
- Android Mobile Browsers (Responsive Interface)

These environments define the supported platforms for deploying and accessing the application.

3.10 Chapter Summary

This chapter presented the Software Requirement Specification for the **College Placement Management System**. It described the purpose, scope, intended audience, user roles, functional requirements, non-functional requirements, and system environment. These requirements establish the foundation for the architectural design and implementation discussed in the following chapters.

ARCHITECTURE

4.1 Introduction

Software architecture defines the overall structure of a system by identifying its major components, their responsibilities, and the interactions between them. A well-designed architecture improves scalability, maintainability, security, and performance while simplifying future enhancements.

The **College Placement Management System** follows a modular three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. This layered approach separates user interface, business logic, and database operations, allowing each layer to function independently while communicating through well-defined interfaces. Such an architecture reduces system complexity and facilitates easier maintenance and future expansion.

4.2 System Architecture Overview

The proposed system provides a centralized platform where different users interact according to their assigned roles. Students, Placement Administrators, Recruiters, and Super Administrators access the application through a web browser. User requests are processed by the backend services, which execute business logic, validate user permissions, interact with the database, and return the required information to the frontend.

The architecture ensures secure communication through JWT-based authentication, enabling only authenticated users to access protected resources. RESTful APIs act as the communication bridge between the frontend and backend, allowing efficient exchange of data while maintaining loose coupling between system components.

4.3 Three-Tier Architecture

The application follows a three-tier architecture consisting of the following layers:

Presentation Layer

The Presentation Layer provides the graphical user interface through which users interact with the application. It is responsible for displaying information, collecting user input, validating client-side data, and communicating with backend services using REST APIs.

This layer is developed using **React**, which enables the creation of reusable user interface components and responsive web pages. Navigation between different modules is managed through client-side routing, providing users with a smooth and interactive experience.

Application Layer

The Application Layer contains the core business logic of the system. It processes user requests, validates business rules, performs eligibility verification, manages placement activities, handles authentication, and communicates with the database layer.

This layer is implemented using **Spring Boot**, which provides a robust framework for developing RESTful web services. The backend also manages role-based authorization, JWT token validation, and interaction with various system modules.

Data Layer

The Data Layer is responsible for storing and managing application data. It maintains information related to students, recruiters, placement drives, applications, notifications, reports, and user accounts.

The application uses **MySQL** as the relational database management system. Database operations are performed through the backend using JPA/Hibernate, ensuring efficient data retrieval, storage, and integrity.

4.4 Technology Stack

The following technologies were used for the development of the proposed system.

Component	Technology Used
Frontend	React
Backend	Spring Boot 3
Programming Language	Java 21
Database	MySQL
Authentication	JWT
Build Tool	Maven
Version Control	Git
Communication	RESTful APIs
Development Environment	Visual Studio Code, IntelliJ IDEA (or Eclipse), MySQL Workbench

The selected technology stack provides high performance, modularity, maintainability, and compatibility with modern web application development practices. The core technologies align with those specified in the SRS.

4.5 System Modules

The College Placement Management System consists of several interconnected modules, each responsible for a specific set of functionalities.

Authentication Module

Provides secure login, logout, password management, JWT generation, and role-based authorization.

Student Management Module

Allows Placement Administrators to create and manage student accounts while enabling students to maintain personal and academic information.

Recruiter Management Module

Facilitates recruiter registration, company profile management, account maintenance, and recruiter access control.

Placement Drive Management Module

Supports creation of placement drives, scheduling of recruitment activities, publication of opportunities, and management of recruitment workflows.

Eligibility Verification Module

Automatically evaluates student eligibility based on criteria such as CGPA, department, graduation year, and company-specific requirements.

Job Application Module

Allows eligible students to browse placement opportunities, submit applications, and monitor application status.

Notification Module

Generates announcements, interview notifications, application updates, and recruitment result notifications for users.

Reporting Module

Generates placement statistics, recruiter reports, student reports, and analytical summaries for administrative decision-making. These modules correspond to the functional features defined in the SRS.

4.6 Data Flow

The overall workflow of the system is as follows:

1. The user accesses the application through a web browser.
2. The user enters valid login credentials.
3. The backend authenticates the user and generates a JWT token.
4. The authenticated user is redirected to the appropriate dashboard based on their role.
5. User requests are sent to the backend through REST APIs.
6. The backend validates permissions and executes the required business logic.
7. Database operations are performed whenever necessary.
8. The processed response is returned to the frontend and displayed to the user.

This workflow ensures secure communication, proper authorization, and efficient processing of placement-related activities.

4.7 Security Architecture

Security is one of the primary considerations in the design of the proposed system.

The application implements **JWT-based authentication** to provide secure, stateless communication between the frontend and backend. Each authenticated user receives a signed JWT token, which must accompany subsequent requests to access protected resources.

Role-Based Access Control (RBAC) ensures that users can access only the functionalities assigned to their respective roles. Passwords are securely stored using encryption techniques, while protected APIs reject requests containing invalid, expired, or missing authentication tokens.

The system architecture also follows REST API standards and incorporates security practices intended to protect user information and application resources.

4.8 Advantages of the Adopted Architecture

The adopted architecture offers several advantages:

- Separation of presentation, business logic, and data management.
- Improved maintainability through modular development.
- Easier debugging and testing due to clearly defined layers.
- Enhanced security using JWT authentication and RBAC.
- Scalable architecture capable of supporting increasing numbers of users.
- Efficient communication through RESTful APIs.
- Simplified future enhancement and module integration.
- Better performance through optimized database interactions.

These architectural characteristics make the system suitable for institutional deployment and future expansion.

4.9 Chapter Summary

This chapter described the architecture of the **College Placement Management System**, including its three-tier architecture, technology stack, system modules, data flow, and security mechanisms. The architecture provides a modular, secure, and scalable foundation for implementing the functional requirements identified in the Software Requirement Specification. The following chapter presents the detailed system design, including database design, UML diagrams, and module-level organization.

SYSTEM DESIGN

5.1 Introduction

System design is the process of transforming software requirements into a structured solution that can be implemented efficiently. It provides a detailed description of the system's architecture, modules, database structure, user interactions, and component relationships. A well-designed system improves maintainability, scalability, and overall software quality by defining clear responsibilities for each module.

The College Placement Management System has been designed using a modular approach in which each functional component performs a specific task while interacting seamlessly with other modules. The design emphasizes separation of concerns, secure communication, efficient data management, and ease of maintenance. The system supports multiple user roles with clearly defined permissions, ensuring secure and organized operation throughout the placement process.

5.2 Module Design

The application is divided into multiple functional modules, each responsible for a particular aspect of the placement process.

Authentication Module

The Authentication Module manages user login, logout, password verification, and role-based authorization. JWT-based authentication is implemented to ensure secure access to protected resources. After successful authentication, users are redirected to dashboards corresponding to their assigned roles.

Student Management Module

This module manages student-related information. Students can update their personal details, academic information, technical skills, certifications, and resumes. Placement Administrators can create, modify, activate, or deactivate student accounts whenever necessary.

Recruiter Management Module

The Recruiter Management Module maintains company information and recruiter accounts. Recruiters can manage their company profiles and participate in placement activities, while Placement Administrators supervise recruiter registrations and account management.

Placement Drive Management Module

This module allows Placement Administrators to create and manage placement drives by specifying company information, job roles, eligibility criteria, application deadlines, interview schedules, and recruitment status.

Eligibility Verification Module

The Eligibility Verification Module automatically checks whether students satisfy the eligibility criteria specified for a placement drive. The verification process considers parameters such as department, academic performance, graduation year, and other recruitment requirements before permitting application submission.

Application Management Module

This module enables eligible students to apply for placement drives. Recruiters can review submitted applications, shortlist candidates, schedule interviews, and publish recruitment outcomes. Students can monitor the progress of their applications throughout the recruitment process.

Notification Module

The Notification Module distributes important placement-related information, including placement announcements, interview schedules, application status updates, and recruitment results. Notifications ensure timely communication among all stakeholders.

Reporting Module

The Reporting Module generates reports related to placement drives, recruiter activities, student participation, placement statistics, and recruitment outcomes. These reports assist administrators in monitoring placement performance and making informed decisions.

5.3 Database Design

The database serves as the central repository for storing application data. A relational database model has been adopted to maintain consistency, integrity, and efficient retrieval of information.

The major entities maintained in the database include:

- User
- Student
- Recruiter
- Company
- Placement Drive
- Eligibility Criteria
- Job Application
- Notification
- Report

These entities are interconnected through appropriate relationships that ensure data consistency while reducing redundancy. The relational design supports efficient execution of placement-related operations such as eligibility verification, application processing, and report generation. The database structure is designed to accommodate future enhancements without affecting existing functionality.

5.4 UML Design

Unified Modelling Language (UML) diagrams were used during the system design phase to visualize system structure, user interactions, and data flow. These diagrams provide a clear understanding of the application's behaviour before implementation.

The following UML diagrams have been prepared for the project:

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram

These diagrams describe the interaction between users and the system, the relationships among classes, the sequence of operations performed during various processes, and the overall workflow of placement activities. They serve as important design documents and facilitate communication among developers and stakeholders.

5.5 User Interface Design

The user interface has been designed with the objective of providing a simple, responsive, and intuitive experience for all users. Separate dashboards are available for each user role to ensure that users can access only the features relevant to their responsibilities.

The interface follows a consistent design across all modules. Forms include appropriate validation to reduce input errors, while navigation menus provide quick access to different functionalities. The responsive frontend ensures compatibility across various screen sizes and modern web browsers.

Key interface components include:

- Login Page
- Super Administrator Dashboard
- Placement Administrator Dashboard
- Recruiter Dashboard
- Student Dashboard
- Placement Drive Management Page
- Student Profile Page
- Job Application Page
- Notification Panel
- Reports Dashboard

The interface design prioritizes usability, accessibility, and efficient navigation, enabling users to perform placement-related tasks with minimal effort.

5.6 Input Design

Input design focuses on collecting accurate and validated information from users.

Major input forms include:

- Login Form
- Student Account Registration Form
- Student Profile Form
- Recruiter Account Creation Form
- Placement Drive Creation Form
- Eligibility Criteria Form
- Job Application Form
- Company Information Form

Each input form validates mandatory fields, data formats, and business rules before submitting information to the backend. Proper validation reduces incorrect data entry and improves overall data quality.

5.7 Output Design

Output design defines how processed information is presented to users.

The system generates various outputs, including:

- Student Dashboard
- Recruiter Dashboard
- Placement Drive Listings
- Eligible Student Lists
- Application Status
- Interview Schedule
- Recruitment Results
- Placement Reports
- Notification Messages

Outputs are displayed in an organized manner to improve readability and support effective decision-making by different stakeholders.

5.8 Design Advantages

The adopted system design provides several benefits:

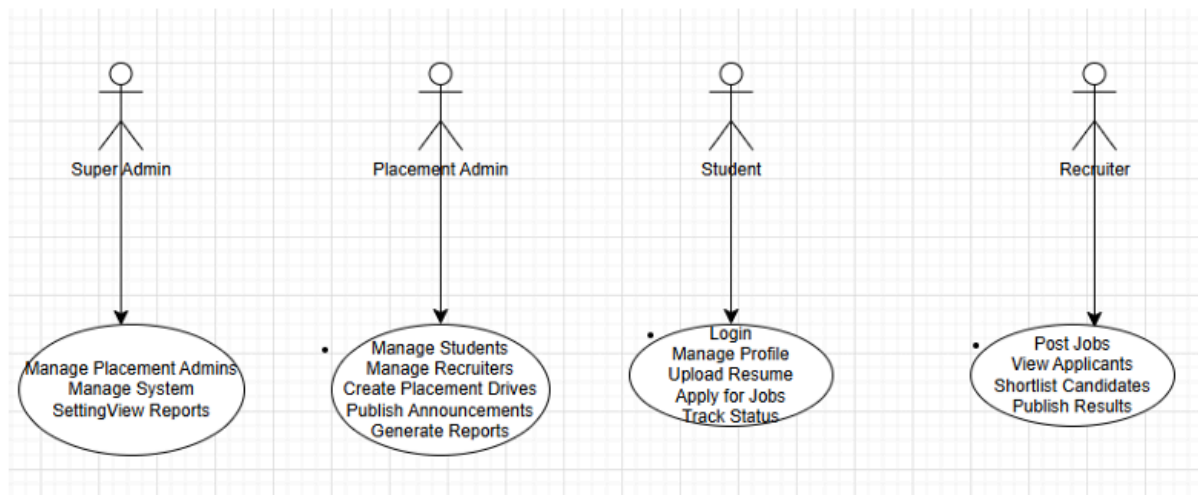
- Modular organization simplifies development and maintenance.
- Clear separation between frontend, backend, and database improves system reliability.
- Standardized interfaces facilitate communication among components.
- Centralized database reduces redundancy and maintains data integrity.
- Role-based design enhances application security.
- Flexible architecture supports future feature additions.
- Reusable modules improve software maintainability.
- Responsive interface enhances user experience across different devices.

5.9 Chapter Summary

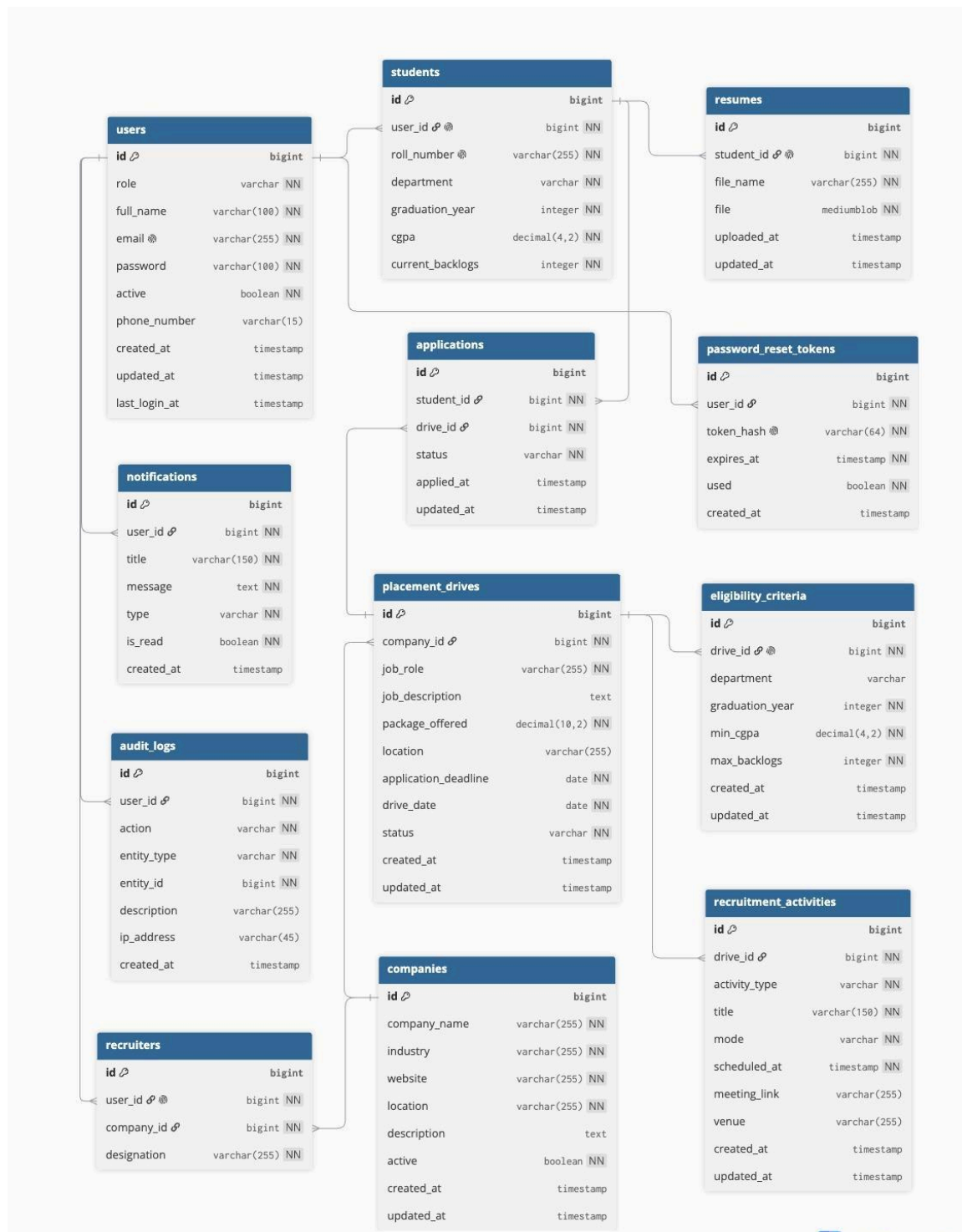
This chapter described the design of the College Placement Management System, including module organization, database design, UML diagrams, user interface design, input and output design, and the advantages of the adopted design approach. The system design provides a structured framework for implementing the application while ensuring scalability, maintainability, and security. The next chapter discusses the implementation of the proposed system and the technologies used during development.

5.10 Diagrams

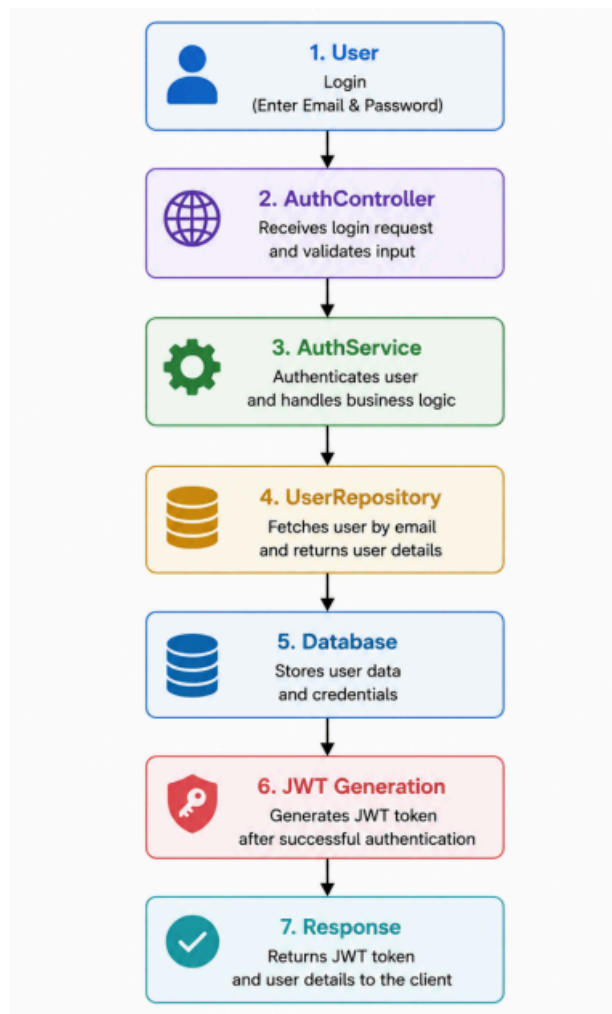
5.10.1 USE CASE DIAGRAM



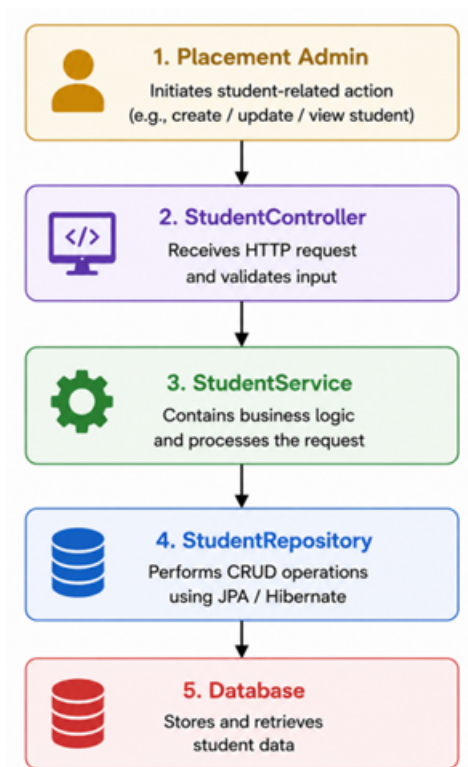
5.10.2 ENTITY RELATIONSHIP DIAGRAM



5.10.3 Sequence diagram



5.10.4 Placement Admin Sequence diagram



IMPLEMENTATION

6.1 Introduction

Implementation is the phase in which the software design is transformed into a functional application. During this stage, all the modules identified in the Software Requirement Specification and System Design are developed, integrated, and tested to ensure that the system performs according to the specified requirements.

The **College Placement Management System** has been implemented as a full-stack web application using modern web technologies. The frontend is developed using React to provide an interactive and responsive user interface, while the backend is implemented using Spring Boot to handle business logic and RESTful web services. MySQL is used as the relational database management system for storing application data, and JWT (JSON Web Token) is employed to provide secure authentication and role-based authorization. The implementation follows the modular architecture described in the previous chapters, ensuring scalability, maintainability, and efficient communication among system components.

6.2 Development Environment

The development of the application was carried out using the following software tools and technologies.

Component	Technology Used
Programming Language	Java 21
Frontend	React
Backend	Spring Boot 3
Database	MySQL
Authentication	JWT
Build Tool	Maven
Version Control	Git
API Testing	Postman
Database Tool	MySQL Workbench
IDE	Visual Studio Code, IntelliJ IDEA (or Eclipse)

These tools provided a reliable and efficient environment for developing, testing, and maintaining the application.

6.3 Project Structure

The project is organized into separate frontend and backend modules, enabling independent development and maintenance of each layer.

Frontend

The frontend is developed using React and is responsible for presenting the user interface. It consists of reusable components, pages, routing, API integration, and responsive layouts.

The frontend includes modules such as:

- Login Interface
- Dashboard
- Student Management
- Recruiter Management
- Placement Drive Management
- Application Management
- Notification Management
- Reports

Backend

The backend is implemented using Spring Boot and follows a layered architecture.

The major backend packages include:

- Controller Layer
- Service Layer
- Repository Layer
- Entity Layer
- Security Configuration
- Exception Handling
- Utility Classes

The backend exposes RESTful APIs that are consumed by the React frontend. These APIs process user requests, perform business logic, and interact with the MySQL database.

6.4 Database Implementation

The application uses MySQL to store all placement-related information in a centralized database.

The database contains tables for:

- Users
- Students
- Recruiters
- Companies
- Placement Drives
- Applications
- Notifications
- Reports

Relationships among these tables are maintained using primary keys and foreign keys, ensuring data integrity and minimizing redundancy. CRUD (Create, Read, Update, Delete) operations are performed using Spring Data JPA, which simplifies interaction with the database and improves maintainability.

6.5 Implementation of Major Modules

Authentication Module

The authentication module verifies user credentials and provides secure access to the application. After successful login, a JWT token is generated and attached to subsequent requests for authentication. The module also implements role-based access control, ensuring that each user accesses only the functionalities assigned to their role.

Student Management Module

This module allows Placement Administrators to create and manage student accounts. Students can update their profiles by adding personal information, academic records, technical skills, certifications, and resumes. The module ensures that student data remains current and readily available during recruitment activities.

Recruiter Management Module

The Recruiter Management Module enables company representatives to maintain company information and participate in campus recruitment. Recruiters can publish job opportunities, define eligibility criteria, review applications, shortlist candidates, and update recruitment status.

Placement Drive Management Module

Placement Administrators can create placement drives by specifying company details, job profiles, eligibility criteria, important dates, and recruitment schedules. The module enables centralized management of all placement activities within the institution.

Eligibility Verification Module

The system automatically evaluates student eligibility based on predefined criteria such as department, academic performance, graduation year, and other company-specific requirements. Only eligible students are allowed to submit applications for placement drives.

Application Management Module

Eligible students can browse available placement drives and submit online applications. Recruiters can review applications, shortlist candidates, and update recruitment stages. Students can track the status of their applications throughout the recruitment process.

Notification Module

The notification module provides timely communication regarding placement announcements, interview schedules, application updates, and recruitment results. Notifications improve coordination among students, recruiters, and placement administrators.

Reporting Module

The reporting module generates analytical reports related to placement drives, recruiter participation, student applications, and placement statistics. These reports assist administrators in monitoring institutional placement performance and planning future recruitment activities.

6.6 API Communication

Communication between the frontend and backend is achieved using RESTful APIs. The frontend sends HTTP requests to backend endpoints, which process the requests and return appropriate responses.

The application makes use of standard HTTP methods:

- **GET** – Retrieve data
- **POST** – Create new records
- **PUT** – Update existing records
- **DELETE** – Remove records

JWT tokens are included in authenticated requests to ensure secure communication between the client and the server.

6.7 Security Implementation

Security has been incorporated throughout the implementation phase.

The application provides:

- JWT-based authentication
- Role-Based Access Control (RBAC)
- Password encryption
- Protected REST APIs
- Session-independent authentication
- Input validation
- Exception handling for unauthorized access

These security mechanisms help safeguard sensitive user information and protect the application from unauthorized access.

6.8 User Interface Implementation

The user interface has been implemented using React with a focus on responsiveness, usability, and consistency. Separate dashboards have been developed for each user role to ensure that users can efficiently access the functionalities relevant to them.

Major interfaces include:

- Login Page
- Super Administrator Dashboard
- Placement Administrator Dashboard
- Recruiter Dashboard
- Student Dashboard
- Placement Drive Management
- Student Profile
- Recruiter Profile
- Reports Dashboard
- Notification Panel

The interface follows a consistent layout with intuitive navigation, enabling users to perform placement-related tasks efficiently.

6.9 Testing During Implementation

During implementation, individual modules were verified to ensure correct functionality before integration. The integrated application was then tested to confirm that communication among modules, database operations, authentication, and role-based authorization worked as expected.

Testing activities included:

- Login verification
- User role validation
- Student profile management
- Placement drive creation
- Eligibility verification
- Job application processing
- Notification generation
- Report generation

These tests confirmed that the implemented modules functioned according to the specified requirements.

6.10 Chapter Summary

This chapter described the implementation of the **College Placement Management System**, including the development environment, project structure, database implementation, major functional modules, API communication, security implementation, user interface, and implementation-level testing. The integration of React, Spring Boot, MySQL, and JWT resulted in a secure, modular, and scalable web application capable of managing campus placement activities efficiently.

CONCLUSION

7.1 Conclusion

The **College Placement Management System** was developed to provide a centralized, secure, and efficient solution for managing campus placement activities within educational institutions. The project successfully addresses the challenges associated with traditional placement management methods by automating various administrative and recruitment processes through a web-based platform.

The application integrates multiple functional modules, including user authentication, student management, recruiter management, placement drive management, eligibility verification, online job applications, notification services, and report generation. These modules work together to simplify the placement process while ensuring accuracy, transparency, and efficient communication among all stakeholders.

The use of **React** for frontend development, **Spring Boot** for backend services, **MySQL** for database management, and **JWT** for secure authentication resulted in a scalable and maintainable application capable of supporting multiple user roles through role-based access control. The modular architecture adopted during development allows different components of the system to operate independently while maintaining seamless integration.

The implemented system significantly reduces manual effort by automating repetitive tasks such as eligibility verification, application management, and report generation. Students can conveniently maintain their profiles, apply for eligible placement drives, and monitor their application status. Recruiters can efficiently publish recruitment opportunities, review applications, shortlist candidates, and update recruitment outcomes. Placement Administrators can monitor the overall placement process through centralized dashboards and analytical reports.

The project successfully satisfies the functional and non-functional requirements defined in the Software Requirement Specification. The resulting application provides a reliable, secure, and user-friendly platform that improves the efficiency of placement management and enhances the overall campus recruitment experience.

7.2 Future Scope

Although the current implementation successfully fulfils the primary objectives of the project, several enhancements can be incorporated in future versions to further improve the functionality and user experience of the system.

Some possible future enhancements include:

- Integration with external job systems to provide students with additional employment opportunities.
- AI-based resume analysis and job recommendation features to improve candidate-job matching.
- Automated interview scheduling with calendar integration.

- Real-time communication through in-application messaging or chatbot support.
- Mobile application development for Android and iOS platforms.
- Advanced analytics and visualization dashboards for placement performance.
- Integration with institutional ERP and Student Information Systems.
- Support for online aptitude tests and technical assessments within the system.
- Email and SMS notification enhancements for improved communication.
- Cloud deployment for higher scalability, availability, and disaster recovery.

These enhancements can further strengthen the system and enable educational institutions to manage increasingly complex placement activities with greater efficiency.

7.3 Final Remarks

The **College Placement Management System** demonstrates how modern web technologies can be effectively utilized to automate and simplify campus recruitment activities. The project provides a practical solution that improves coordination among students, recruiters, and placement administrators while ensuring secure data management and efficient workflow automation.

The modular architecture, secure authentication mechanism, responsive user interface, and centralized database collectively contribute to a reliable and maintainable software solution. The knowledge gained during the analysis, design, development, and implementation phases of this project has provided valuable practical experience in full-stack web application development and software engineering principles.

Overall, the project successfully achieves its intended objectives and establishes a strong foundation for future enhancements and wider institutional adoption.

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